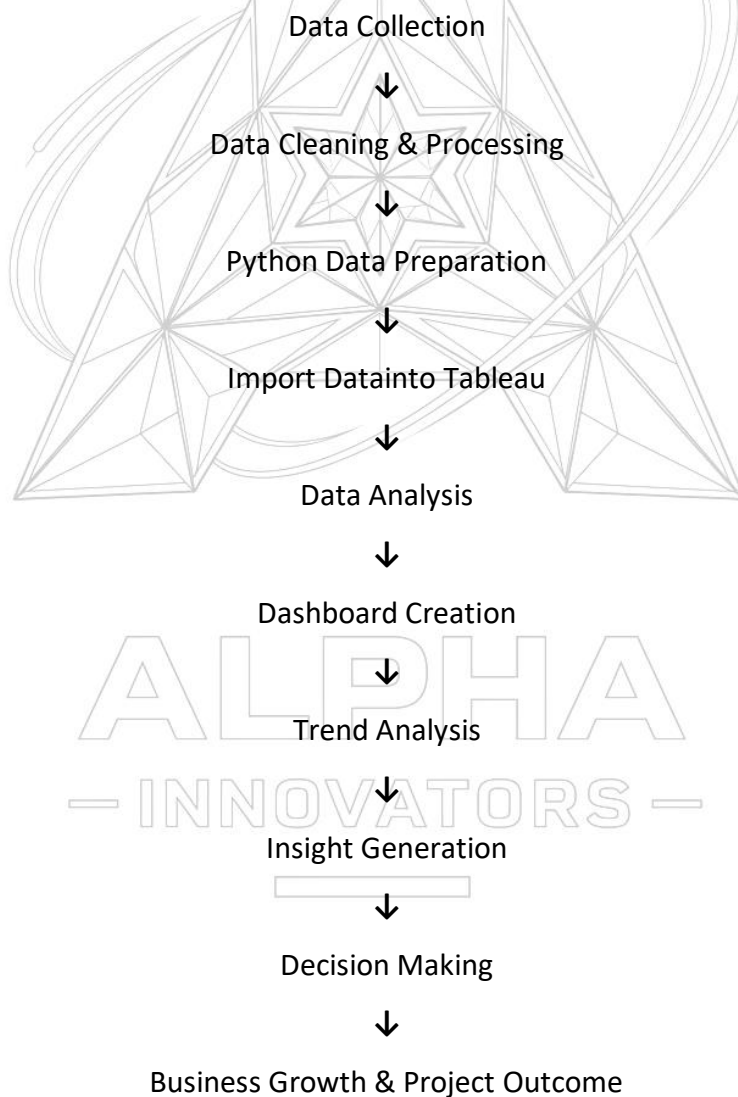


COSMETIC INSIGHTS: Navigating Cosmetics Trends and Consumer Insights With Tableau

TEAM ID	SWTID-2026-8806
PROJECT TITLE	COSMETIC INSIGHTS: Navigating Cosmetics Trends and Consumer Insights With Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture



COSMETIC INSIGHTS: Navigating Cosmetics Trends and Consumer Insights With Tableau

The project "Cosmetic Insights" was created to examine and interpret various types of data related to the cosmetic industry and give insight to business decisions through Tableau; the architecture to accomplish these tasks includes several stages of data collection, processing, analysis, and visualization to facilitate making strategic decisions.

Data Sources:The first stage of the project is to collect different types of data from many different sources, including but not limited to, cosmetic product datasets, sales & revenue records, customer reviews, e-commerce sites, social media data, and consumer demographic data, all of which will ultimately be used for analysis.

Data Ingestion:All types of collected data will be ingested into the system using CSV files, Excel sheets and database connections. At this stage of the project, all collected data from different sources is combined into a single process workflow.

Data Processing:The processing stage of the project includes cleaning the data, transforming raw-format data into structured-format data, and ensuring data quality. Cleaned and processed data will also have duplicate records removed, missing/invalid value corrections, transformed into structured format (although not limited to: service-level agreements) standardization of information for product classifications & sales prices, and other standard formatting corrections will be accomplished during this phase.

Data Storage:After the completion of the processing stage, all structured-format data will be stored in either CSV files, Excel sheets, or MySQL databases with security provisions for data preservation and/or retrieval for analysis.